

BSIT 375: Knowledge Discovery and Data Science

Week 5 Lecture Notes: Building Modern Data Pipelines with AWS Services From Ingestion to Machine Learning

Instructor: Tek Raj Pant

April 2026

Abstract

Modern data science goes beyond algorithms; it requires robust, scalable data pipelines. This lecture introduces the structure of a data pipeline and covers key AWS services for each stage: storage (S3), real-time ingestion (Kinesis, SQS), ETL and processing (Glue, Apache Spark), querying (Athena), warehousing (Redshift), and machine learning (SageMaker). We discuss how data scientists use these tools to build production-ready workflows.

Contents

1	What is a Data Pipeline?	4
1.1	Why Data Pipelines matter in Data Science	4
1.2	Typical Stages of a Data Pipeline	4
2	AWS S3 – The Data Lake Foundation	5
2.1	What is Amazon S3?	5
2.2	How Data Scientists Use S3	5
3	Data Ingestion: Kinesis and SQS	6
3.1	Amazon Kinesis	6
3.2	Amazon SQS (Simple Queue Service)	6

4	ETL Processes and AWS Glue	7
4.1	What is ETL?	7
4.2	AWS Glue	7
4.2.1	Key Components	7
4.2.2	How Data Scientists Use Glue	7
5	Apache Spark in Data Science Pipelines	8
5.1	Spark vs Traditional Python (Pandas)	8
5.2	Spark on AWS	8
5.3	When Data Scientists Use Spark	9
6	Amazon Athena – Serverless Querying on S3	9
6.1	What is Athena?	9
6.2	How It Works	9
6.3	Data Science Use Cases	9
7	Amazon Redshift – Cloud Data Warehousing	10
7.1	What is Redshift?	10
7.2	Key Concepts	10
7.3	Data Scientist’s Role with Redshift	10
7.4	Typical Data Flow with Redshift	10
8	Amazon SageMaker – Machine Learning at Scale	11
8.1	What is SageMaker?	11
8.2	How SageMaker Fits in the Data Pipeline	11
8.3	Data Science Workflow Example	11
8.4	SageMaker and Other AWS Services	11
9	Putting It All Together – An End-to-End Data Science Pipeline	12
9.1	Scenario: E-commerce Clickstream Analysis and Churn Prediction	12
10	Summary and Best Practices	12

1 What is a Data Pipeline?

Definition

A **data pipeline** is an automated, end-to-end process that moves data from source systems to destination systems, applying transformations, validations, and enrichment along the way.

Why Data Pipelines matter in Data Science

- Raw data is rarely ready for analysis – it needs cleaning, joining, and reshaping.
- Pipelines ensure reproducibility, scalability, and monitoring.
- A well-designed pipeline separates storage, compute, and orchestration, allowing each to scale independently.

Typical Stages of a Data Pipeline

1. **Ingestion:** Capture data from sources (databases, IoT devices, logs, APIs).
2. **Storage:** Durable, cost-effective storage (data lakes).
3. **Processing & ETL:** Transform raw data into analytics-ready formats.
4. **Querying & Warehousing:** Enable fast SQL analytics.
5. **Machine Learning:** Train and deploy models.
6. **Orchestration:** Coordinate all steps (not covered in depth today).

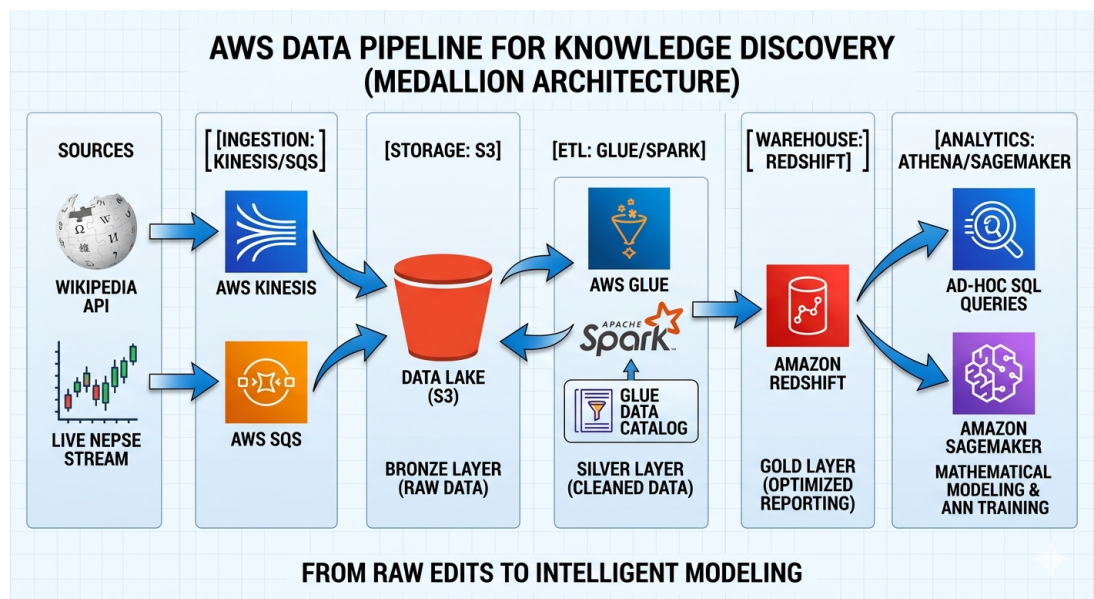
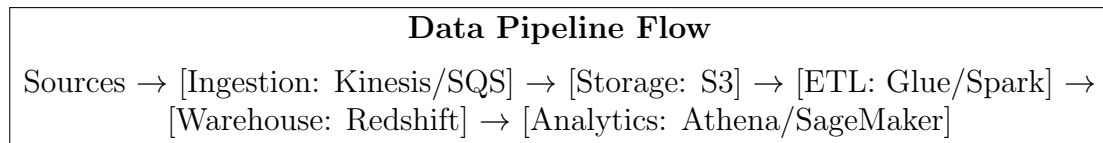


Figure 1: AWS Data Pipeline for Wikipedia Knowledge Discovery (Medallion Architecture)

Figure 2: Conceptual architecture of an AWS-based data pipeline

2 AWS S3 – The Data Lake Foundation

What is Amazon S3?

Simple Storage Service (S3) is an object storage service with high durability (99.999999999%), scalability, and low cost. It is the most common data lake storage for data science.

How Data Scientists Use S3

- Store raw CSV, JSON, Parquet, or Avro files.
- Intermediate and final transformed datasets.
- Model artifacts (e.g., trained models from SageMaker).
- Serve data to Athena, EMR, Glue, or Redshift Spectrum.

Best Practice

Organise S3 buckets with a hierarchical key structure:
s3://my-bucket/raw/2026/04/15/, s3://my-bucket/processed/,
s3://my-bucket/curated/. Use partitioning for query performance.

3 Data Ingestion: Kinesis and SQS

Real-time and near-real-time ingestion are essential for streaming analytics and event-driven pipelines.

Amazon Kinesis

Kinesis is a platform for streaming data ingestion and processing. Key services:

- **Kinesis Data Streams:** Capture and store streaming records (e.g., clickstreams, IoT sensor data).
- **Kinesis Data Firehose:** Automatically load streaming data into S3, Redshift, or Elasticsearch.
- **Kinesis Data Analytics:** Run SQL or Apache Flink on streams.

Data science use case: Real-time feature extraction (e.g., sliding window aggregations) before storing in S3 for batch ML.

Amazon SQS (Simple Queue Service)

- Fully managed message queue for decoupling components.
- Asynchronous communication: a producer sends a message, a consumer processes it.
- Guarantees at-least-once delivery.

Data science use case: Reliably buffer requests to a prediction endpoint, handle sudden load spikes, or coordinate ETL jobs.

Table 1: Kinesis vs SQS – When to use which?

Feature	Kinesis Data Streams	SQS
Message retention	Up to 365 days	Up to 14 days
Processing model	Real-time, replayable, multiple consumers	Standard/FIFO queue, single consumer
Ordering	Within shard	FIFO queue supports ordering
Data size per record	Up to 1 MB	Up to 256 KB (extended: 2 GB)
Best for	Streaming analytics, replay	Decoupling microservices, buffering

4 ETL Processes and AWS Glue

What is ETL?

ETL stands for Extract, Transform, Load. It is the process of:

- **Extract:** Reading data from source(s) – S3, RDS, streaming, etc.
- **Transform:** Cleaning, enriching, joining, aggregating, changing schema.
- **Load:** Writing the result to a target (data warehouse, data lake, database).

AWS Glue

AWS Glue is a serverless data integration service that simplifies ETL.

Key Components:

- **Glue Data Catalog:** Central metadata repository storing table definitions, schemas, and partition information.
- **Glue ETL Jobs:** Apache Spark-based serverless jobs (Python or Scala) that perform transformations.
- **Glue Crawlers:** Automatically scan S3 or databases to populate the Data Catalog.
- **Glue Workflows:** Orchestrate multi-job ETL pipelines.

How Data Scientists Use Glue:

1. Run crawlers on raw S3 data to infer schema and create tables.
2. Write PySpark ETL scripts to clean and aggregate data.
3. Store the transformed data back to S3 (e.g., in Parquet format partitioned by date).
4. Use the Data Catalog to make data available for Athena, Redshift Spectrum, or SageMaker.

Listing 1: Simple Glue PySpark ETL snippet

```
import sys
from awsglue.transforms import *
from awsglue.context import GlueContext
from pyspark.context import SparkContext
from awsglue.dynamicframe import DynamicFrame
```

```

sc = SparkContext()
glueContext = GlueContext(sc)

# Read from Data Catalog
dyf = glueContext.create_dynamic_frame.from_catalog(
    database="my_db",
    table_name="raw_clickstream"
)

# Transform: filter and cast
dyf = dyf.filter(lambda row: row['event_type'] == 'purchase')
dyf = dyf.rename_field('user_id', 'customer_id')

# Write to S3 as Parquet
glueContext.write_dynamic_frame.from_options(
    frame=dyf,
    connection_type="s3",
    connection_options={"path": "s3://my-bucket/clean/purchases/"},
    format="parquet"
)

```

5 Apache Spark in Data Science Pipelines

Definition

Apache Spark is a unified analytics engine for large-scale data processing. It provides APIs in Python, Scala, Java, and R, and includes modules for SQL, streaming, machine learning (MLlib), and graph processing (GraphX).

Spark vs Traditional Python (Pandas)

- Pandas works in-memory on a single machine; Spark distributes data across a cluster.
- Spark can handle terabytes or petabytes; Pandas is limited by RAM.
- For data science, Spark is ideal for data preparation at scale; then you can sample down for local modelling.

Spark on AWS

AWS offers several ways to run Spark:

- **AWS Glue** (serverless Spark – no cluster management).

- **Amazon EMR** (Elastic MapReduce) – full control over Spark, Hadoop, Hive.
- **SageMaker Notebooks** with SparkMagic or EMR integration.

When Data Scientists Use Spark

1. Feature engineering on massive datasets (e.g., aggregations over user sessions).
2. Preprocessing before training distributed ML models (e.g., using Spark MLlib).
3. Reading from S3, joining multiple tables, writing back.

6 Amazon Athena – Serverless Querying on S3

What is Athena?

Athena is a serverless, interactive query service that lets you analyse data directly in S3 using standard SQL. You pay only for the amount of data scanned.

How It Works

- Athena uses Presto under the hood.
- It relies on the Glue Data Catalog for schema information.
- Supports formats: CSV, JSON, Parquet, Avro, ORC.

Data Science Use Cases

- Quickly explore raw data without ETL.
- Run ad-hoc SQL queries to understand distributions, missing values, etc.
- Validate output of ETL jobs.
- Build dashboards (via QuickSight) on S3 data.

Optimisation Tip

Store data in columnar formats like Parquet or ORC and partition by commonly filtered columns (e.g., `year/month/day`). This reduces scanned data and lowers costs.

Listing 2: Example Athena query on partitioned Parquet data

```
SELECT device_type , COUNT(*) AS total_events
FROM my_db.clickstream
WHERE year = '2026' AND month = '04'
GROUP BY device_type;
```

7 Amazon Redshift – Cloud Data Warehousing

What is Redshift?

Amazon Redshift is a fast, petabyte-scale, fully managed cloud data warehouse. It uses columnar storage and massively parallel processing (MPP).

Key Concepts

- **Cluster:** Set of leader and compute nodes.
- **Columnar storage:** Each column stored separately, improves I/O for analytical queries.
- **Distribution style and sort keys:** Optimise query performance.
- **Redshift Spectrum:** Query exabytes of data in S3 without loading.

Data Scientist's Role with Redshift

- Connect from Python (e.g., `psycopg2`, `sqlalchemy`) to pull aggregated datasets.
- Run complex SQL that would be slow on Athena (Redshift is a dedicated warehouse).
- Use Redshift ML to train models using SQL (`CREATE MODEL`) – experimental but promising.
- Usually, ETL pipelines (Glue/Spark) load curated tables into Redshift for business reporting.

Typical Data Flow with Redshift

Raw data in S3 -> Glue ETL (Spark) -> Cleaned data in S3 ->
Redshift COPY command -> Redshift tables ->
Queries from BI tools / SageMaker

8 Amazon SageMaker – Machine Learning at Scale

What is SageMaker?

Amazon SageMaker is a fully managed service that covers the entire ML lifecycle: data labeling, feature engineering, training, tuning, deployment, and monitoring.

How SageMaker Fits in the Data Pipeline

1. **Data preparation:** SageMaker Data Wrangler (visual interface) or Processing Jobs (running Spark or sklearn).
2. **Feature store:** SageMaker Feature Store – centralised repository for features shared across teams.
3. **Training:** Managed training instances, automatic model tuning, distributed training.
4. **Deployment:** Real-time endpoints, asynchronous inference, batch transform.
5. **Pipeline orchestration:** SageMaker Pipelines (ML-specific CI/CD).

Data Science Workflow Example

- Raw data in S3 (ingested via Kinesis Firehose).
- Glue ETL creates training datasets in Parquet.
- SageMaker Processing uses a Spark container to further engineer features.
- Feature groups registered in Feature Store.
- Training job reads from S3 or Feature Store, produces a model.
- Model deployed to an endpoint; SQS queues requests for asynchronous inference.

SageMaker and Other AWS Services

- **S3:** Input/output for datasets, models, and logs.
- **Kinesis:** Real-time predictions on streaming data (using SageMaker endpoints directly).
- **Glue:** Prepare training data (can also be called from SageMaker Pipelines).
- **Redshift:** Pull data for training (via Redshift Data API).
- **Athena:** Explore data before building models.

9 Putting It All Together – An End-to-End Data Science Pipeline

Let us construct a realistic scenario:

Scenario: E-commerce Clickstream Analysis and Churn Prediction

1. **Ingestion:** User clicks are sent to Kinesis Data Streams in real time.
2. **Storage:** Kinesis Firehose buffers and writes raw JSON to S3 (partitioned by date/hour).
3. **ETL:** Daily Glue Spark job reads the raw S3 data, cleans invalid events, joins with user metadata from a database (via JDBC), and writes aggregated session features as Parquet to S3.
4. **Catalog:** Glue Crawler updates the Data Catalog with new partitions.
5. **Querying:** Data scientists use Athena to run quick exploratory queries on the aggregated features.
6. **Warehouse:** For dashboarding, the Glue job also loads the aggregated data into Redshift.
7. **ML:** SageMaker Notebook (or Processing) reads the Parquet features from S3, trains an XGBoost model for churn, and registers the model in SageMaker Model Registry.
8. **Inference:** The model is deployed to a real-time endpoint. New streaming events (via Kinesis) call the endpoint to predict churn in real time. For heavy batch predictions, the model is run as a SageMaker Batch Transform job on S3 data.
9. **Monitoring:** SageMaker Model Monitor detects data drift; when drift is significant, a pipeline retrains the model automatically.

10 Summary and Best Practices

- A robust data pipeline separates concerns: ingestion, storage, processing, warehousing, and ML.
- **Start with S3** as your data lake – it is the centre of gravity for almost all services.
- Use Kinesis for streaming and SQS for reliable decoupling.
- Prefer Glue (serverless Spark) over managing your own Spark cluster for standard ETL.
- Always use columnar formats (Parquet/ORC) and partitioning to optimise cost and performance.

- Let data scientists explore with Athena, but shift heavy analytics to Redshift.
- SageMaker handles the ML lifecycle – but the quality of your pipeline upstream determines model success.

Table 2: AWS Services and Their Pipeline Stage

Stage	AWS Service	Role in Data Science
Ingestion	Kinesis, SQS	Capture streaming data, buffer events
Storage	S3	Data lake for raw, intermediate, and final datasets
Catalog	Glue Data Catalog	Metadata management, schema discovery
ETL	Glue (Spark), EMR	Transform and prepare features
Query	Athena	Ad-hoc SQL on S3
Warehousing	Redshift	Fast BI and complex SQL on curated data
ML	SageMaker	Training, deployment, feature store, monitoring

11 References and Further Reading

- AWS Documentation: <https://docs.aws.amazon.com>
- Larose, D. T., & Larose, C. D. (2014). *Discovering Knowledge in Data* (2nd ed.). Wiley.
- Apache Spark: <https://spark.apache.org>
- “Data Pipelines Pocket Reference” by James Densmore (O’Reilly).